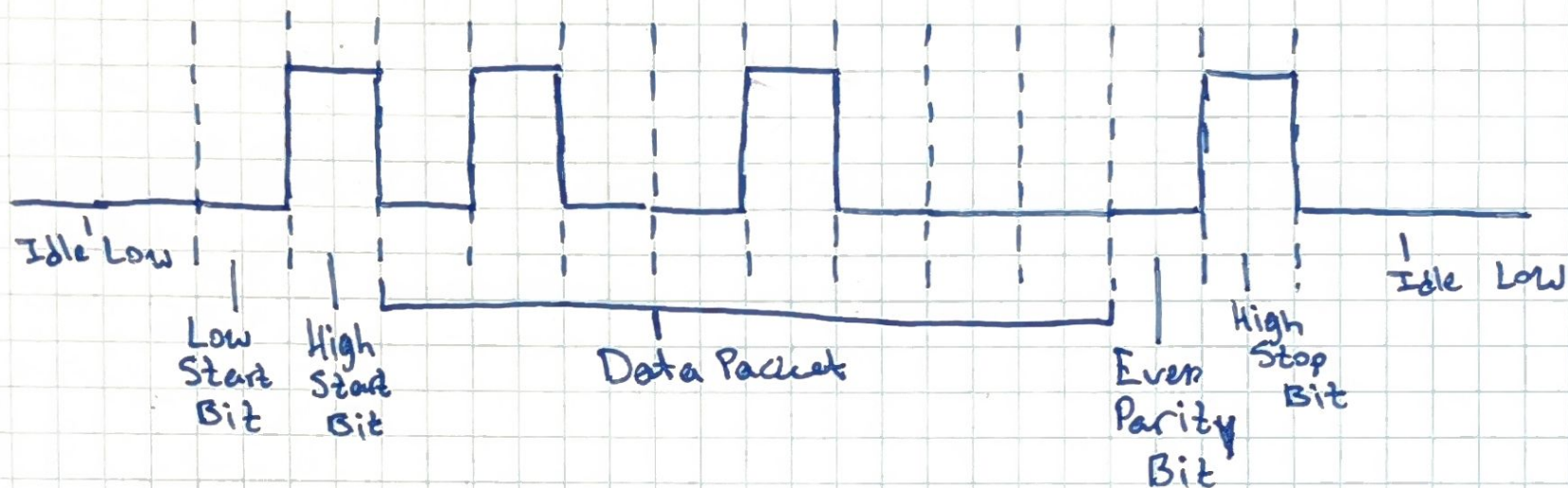


IR Communication Protocol

• Frame Format

- Low Start Bit
- High Start Bit
- Data Packet (8 bits) (LSB First)
- Parity Bit (Even)
- High Stop Bit
- Idle Low

• Example transmission: 18 = $0x12$ = 0b 0001 0010



* Reception is Sampled in the middle of each bit

* High/Low represented by modulating duty cycle of PWM signal

High - Duty Cycle = 50%

Low - Duty Cycle = 0%

* Our Emitters transmit at frequency of 38 kHz

* Transmit time ~562 μ s (~21 cycles/bit) (on ESP32-S3)

~750 μ s (~28 cycles/bit) (on ESP32-C6)